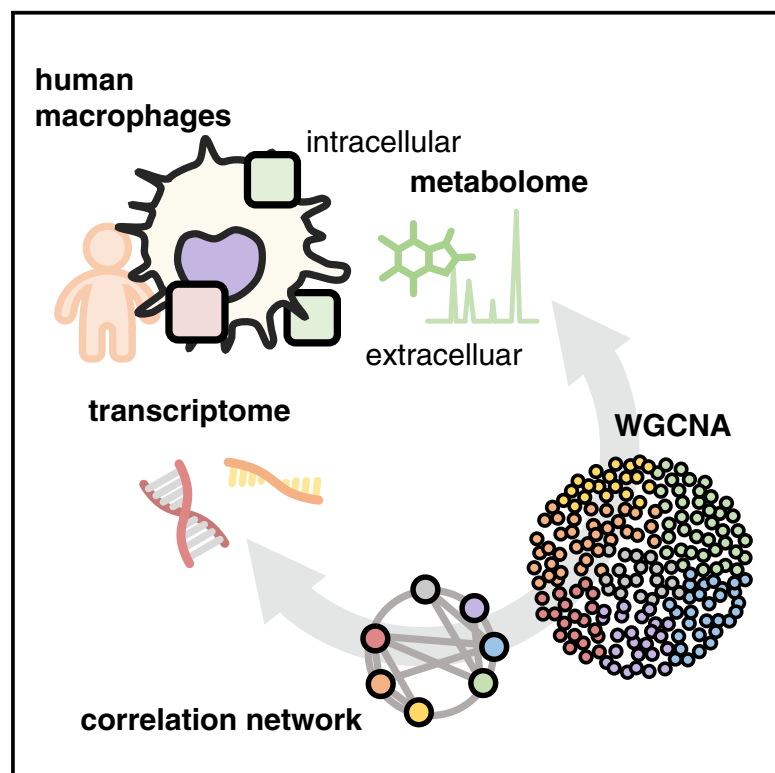


Integrated metabolic-transcriptomic network identifies immunometabolic modulations in human macrophages

Graphical abstract



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In brief

Chen et al. present an extensive multi-omics dataset integrating intra- and extracellular metabolomes with transcriptomic data of human macrophages. Their study reveals several immunometabolic modulations, highlighting key metabolic processes and potential therapeutic targets for immune-related disorders.

Highlights

- Metabolite-gene correlation model identifies immunometabolic alterations bi-directionally
- Tryptophan metabolism characterizes pro-/anti-inflammatory macrophages
- Fc receptor signaling drives glycolysis
- Cholesterol accumulation dampens macrophage interferon responses



Article

Integrated metabolic-transcriptomic network identifies immunometabolic modulations in human macrophages

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SUMMARY

Macrophages exhibit diverse phenotypes and respond flexibly to environmental cues through metabolic re-modeling. In this study, we present a comprehensive multi-omics dataset integrating intra- and extracellular metabolomes with transcriptomic data to investigate the metabolic impact on human macrophage function. Our analysis establishes a metabolite-gene correlation network that characterizes macrophage activation. We find that the concurrent inhibition of tryptophan catabolism by IDO1 and IL411 inhibitors suppresses the macrophage pro-inflammatory response, whereas single inhibition leads to pro-inflammatory activation. We find that a subset of anti-inflammatory macrophages activated by Fc receptor signaling promotes glycolysis, challenging the conventional concept of reduced glycolysis preference in anti-inflammatory macrophages. We demonstrate that cholesterol accumulation suppresses macrophage IFN- γ responses. Our integrated network enables the discovery of immunometabolic features, provides insights into macrophage functional metabolic reprogramming, and offers valuable resources for researchers exploring macrophage immunometabolic characteristics and potential therapeutic targets for immune-related disorders.

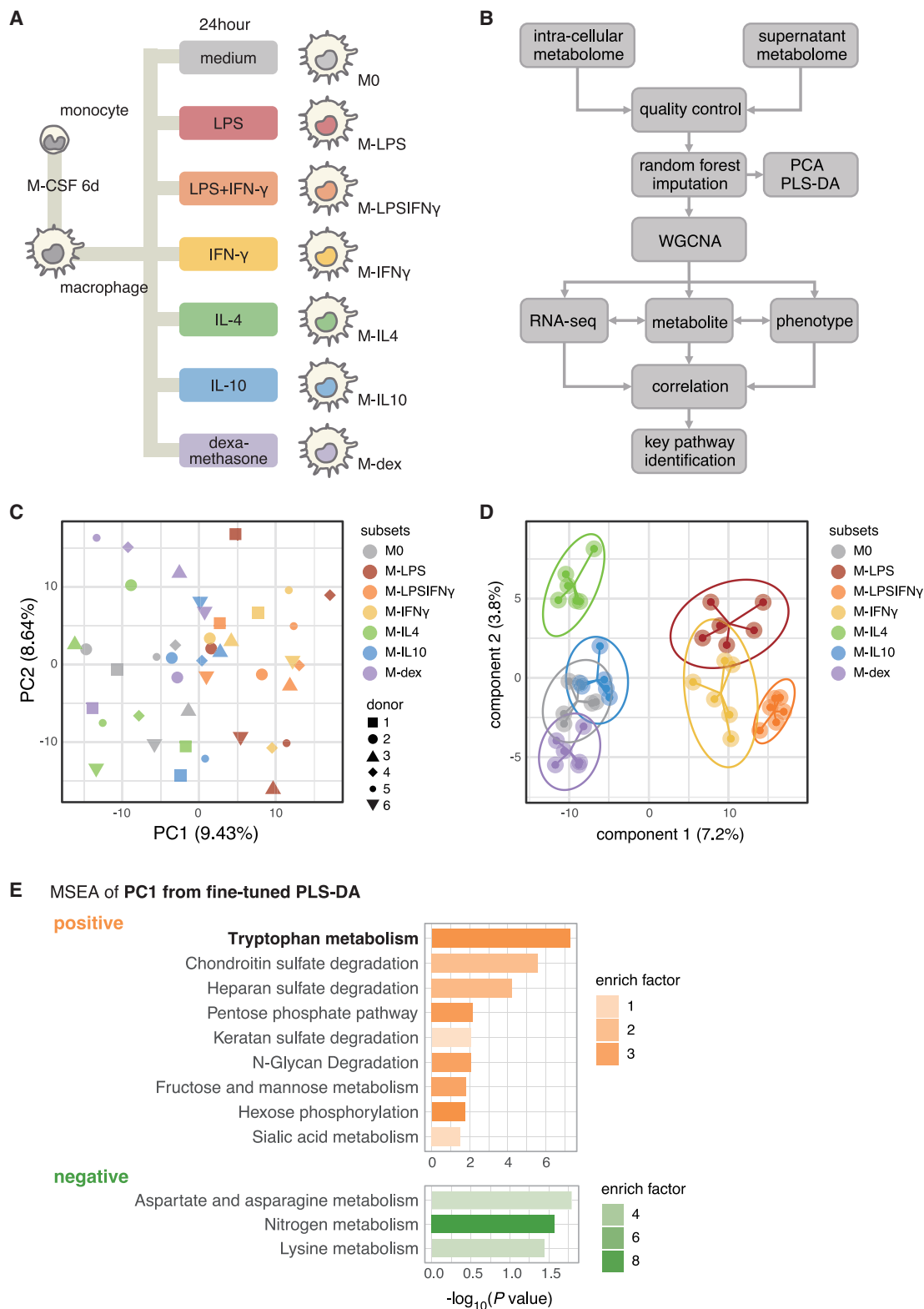
INTRODUCTION

Macrophages are highly plastic innate immune cells that can adapt to environmental cues and perform various functions.^{1,2} The plasticity of tissue resident macrophages, also partly determined by their origin, leads to their diverse roles *in vivo*.³ They can act as pro-inflammatory immune sentinels during pathogen tissue invasion or adopt reparative features for inflammation resolution and wound healing.^{4,5} This delicate homeostasis can be disturbed if the balanced responses are not properly regulated. Dysregulated macrophages are associated with disorders and specific states of disease,⁶ such as COVID-19,^{7,8} atherosclerosis,⁹ autoimmunity,¹⁰ fibrosis,¹¹ and cancer.¹² The classical M1/M2 paradigm simplifies macrophage diversity by categorizing them into pro- and anti-inflammatory states.^{13,14} Researchers have utilized this concept to establish *in vitro* macro-

phage models by activating monocyte-derived macrophages (MDMs) with different stimuli to generate M1s (tumor necrosis factor [TNF], lipopolysaccharide [LPS], interferon- γ [IFN- γ]) and M2s (interleukin-4 [IL-4], IL-10, transforming growth factor- β , glucocorticoids). Transcriptomic studies have identified changes in gene expression that reflect the distinct phenotypes of pro- and anti-inflammatory (M1/M2) macrophages.^{15–17}

Ample evidence supports the critical role of metabolic re-modeling in regulating macrophage responses.^{18–21} Pro-inflammatory macrophages rely primarily on glycolysis as their main energy source, while anti-inflammatory macrophages are more dependent on oxidative phosphorylation (OXPHOS) and fatty acid utilization.^{22,23} However, these generalized concepts are mainly based on mouse studies.²⁴ Additionally, while there is significant heterogeneity within commonly used M2 models,^{15,25,26} most of the immunometabolism studies focus





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on IL-4-activated macrophages. Furthermore, intracellular metabolomics has been the dominant research method for studying macrophage immunometabolism. Recent advances highlight the importance of secreted or environmental metabolites, which can be sensed by different mechanisms, thereby influencing intracellular signaling and intercellular communication.^{27,28} Therefore, comprehensive profiling of both intra- and extracellular metabolome and systematic analysis across different macrophage subsets are essential for the comprehension of the immunometabolic orchestration driving macrophage activation.

In this study, we conducted nontargeted metabolomics on intra- and extracellular samples obtained from seven commonly used human MDM models to explore the immunometabolic orchestration in macrophages. We identified 65 metabolite modules with distinct accumulation patterns and integrated them with transcriptional profiles. Using these systems-level approaches, we discovered several immunometabolic features, including the essential role of intact tryptophan catabolic fluxes in macrophage pro-inflammatory responses, lactate production triggered by Fc receptor (FcR) signaling, and specific dampening of macrophage IFN- γ responses by cellular cholesterol accumulation. Our findings offer valuable resource datasets for the scientific community, providing a comprehensive metabolome-transcriptome profile of human macrophages. This serves as a vital foundation for simulating and predicting macrophage behavior under various conditions, facilitating hypothesis generation and experimental design.

RESULTS

Tryptophan metabolism distinguishes pro-/anti-inflammatory macrophages

We aimed to identify metabolic signatures that distinguish between different macrophage activation states. We employed unstimulated macrophages (M0) and well-established pro- and anti-inflammatory macrophage models¹⁵ by activating MDMs with LPS (M-LPS), LPS and IFN- γ (M-LPSIFN γ), M-IFN γ , M-IL4, M-IL10, and dexamethasone (M-dex) (Figure 1A). We confirmed proper activation by the expression of stimulus-specific pro-/anti-inflammatory markers in these macrophages (Figure S1A). The metabolite ion content of cell lysates and supernatants was analyzed using non-targeted flow-injection analysis-mass spectrometry, generating accurate mass-based tentative metabolite annotations without distinguishing isomers.²⁹ After quality control filtering and random-forest imputation^{30,31} (Figures 1B, S1B, and S1C; see STAR Methods), we obtained 13,418 ions from this combined intra- and extracellular metabolome (Table S1). Among these ions, 5,136 were putatively matched to known metabolites based on their mass-to-charge features (m/z ratios).

Principal-component (PC) analysis revealed a distinct separation of classic pro- and anti-inflammatory activation along PC1. Notably, M0 (non-activated) macrophages distributed at the negative side of PC1, similar to the anti-inflammatory subsets (Figure 1C). This may be explained by the anti-inflammatory nature of M-colony-stimulating factor (CSF)-differentiated macrophages, compared to their granulocyte macrophage-CSF counterpart.^{32,33} Metabolite set enrichment analysis (MSEA) of ions in PC1 showed that metabolites enriched in cytochrome P450 and tryptophan (Trp) metabolism were correlated with M1, while metabolites involved in 3-oxo-10R-octadecatrienoate β -oxidation had negative scores, indicating a more anti-inflammatory phenotype (Figure S1D). For enhanced activation-specific feature selection, we employed partial least-squares-discriminant analysis (PLS-DA; Figures S1E, and S1F), followed by a stability test to ensure robustness and reliability³⁴ (Figure S1G). The fine-tuned PLS-DA confirmed the dichotomy between pro- and anti-inflammatory/M0 macrophages (Figure 1D). MSEA of tuned component 1 confirmed that Trp metabolism was the most consistent and reliable contributor to differentiating classic pro- and anti-inflammatory macrophages (Figure 1E).

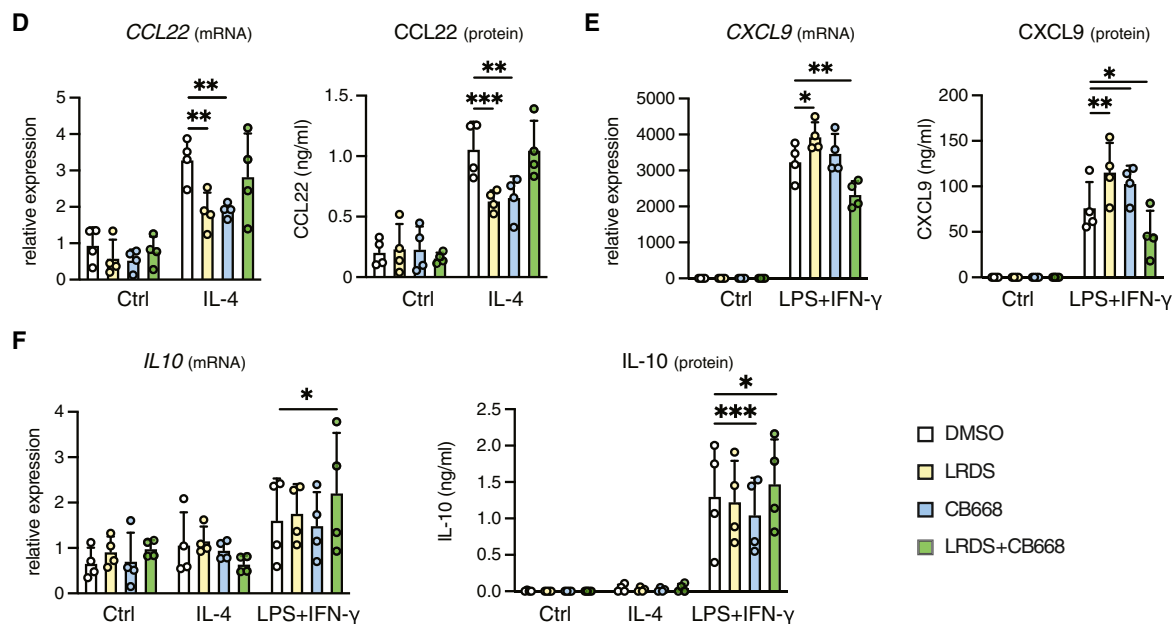
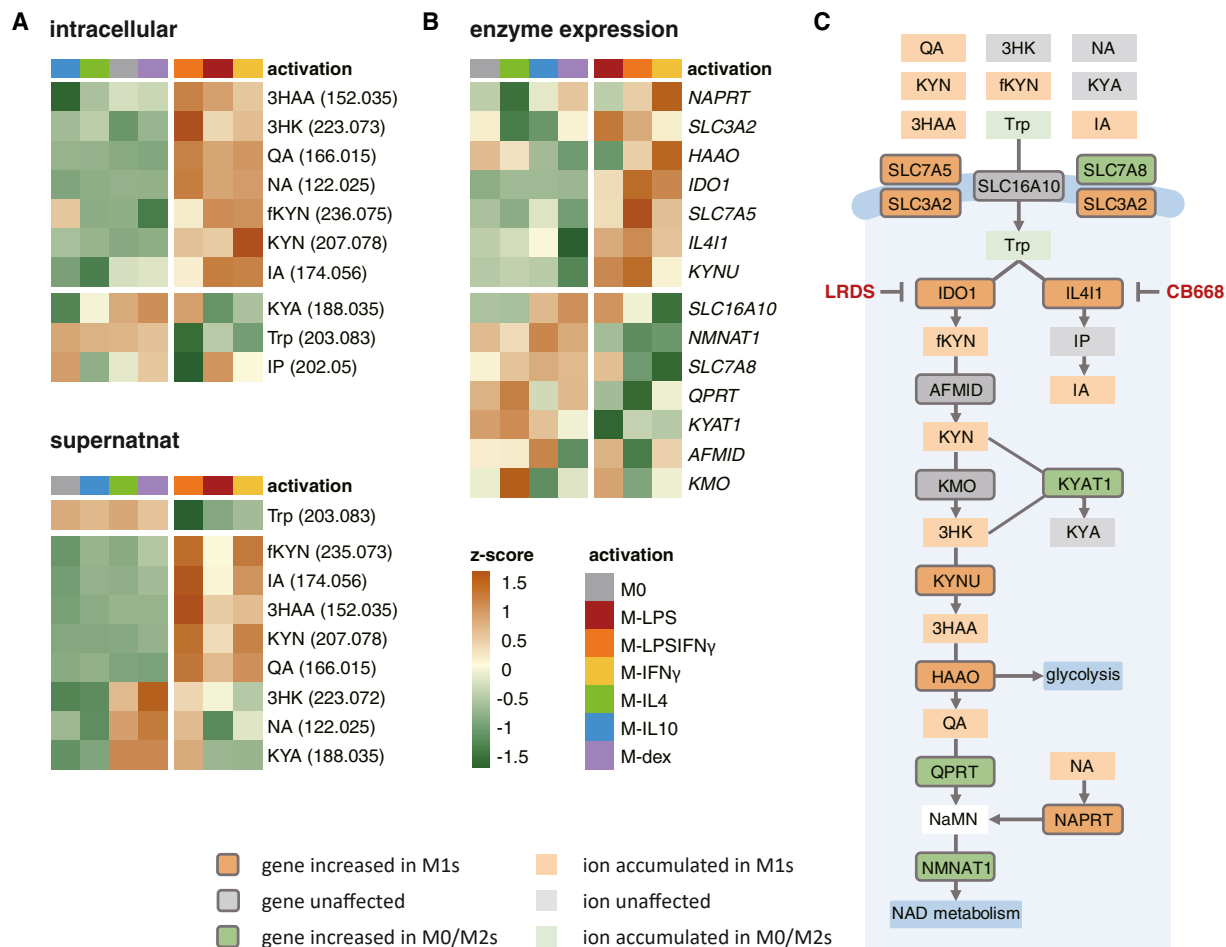
Perturbed Trp metabolism by IDO1 and IL4I1 inhibitors suppresses macrophage inflammation

To investigate the role of Trp metabolism in macrophage activation, we examined metabolite levels (Figures 2A and S2A) and enzyme gene expression (Figures 2B and S2B)¹⁵ related to the Trp catabolic pathway (Figure 2C).³⁵ Our results revealed that pro-inflammatory activation led to the accumulation of Trp catabolites, including kynurenine (KYN), indole acetate (IA), and quinolinic acid (QA), along with a reduction in intracellular and extracellular Trp levels. Pro-inflammatory stimulation upregulated Trp membrane transporters and key catabolic enzymes *IDO1* and *IL4I1*, indicating the involvement of Trp catabolism in pro-inflammatory activation. These findings suggest that Trp catabolism is enhanced in pro-inflammatory macrophages, resulting in increased Trp uptake and consumption.

The impact of inhibiting Trp catabolism on macrophage function remains controversial.^{36,37} Since Trp can be processed via both IDO1- and IL4I1-mediated pathways, we hypothesized that both Trp fluxes are essential for macrophage responses to pro-inflammatory stimulation. To test this, we treated MDMs with IDO1 inhibitor linrodostat (LRDS)³⁸ and/or IL4I1 inhibitor CB668,³⁹ followed by IL-4 or LPS plus IFN- γ stimulation. Interestingly, in M-IL-4, single enzyme inhibition with LRDS or CB668 alone downregulated both the mRNA and protein production of anti-inflammatory marker *CCL22*,^{40,41} while dual inhibition restored its expression (Figure 2D). Other anti-inflammatory markers *MRC1* (CD206) and *CD209* also showed similar patterns (Figure S2C). Conversely, single inhibitor-treated M-LPSIFN γ exhibited increased expression levels and secretion

Figure 1. Tryptophan metabolism differentiates pro- and anti-inflammatory macrophages

- (A) Experimental procedure for generating differentially activated macrophages.
- (B) Analysis workflow.
- (C) PC analysis of macrophage metabolome.
- (D) Tuned PLS-DA.
- (E) Metabolite set enrichment analysis of component 1 from fine-tuned PLS-DA.



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of the pro-inflammatory chemokine CXCL9,^{40,41} whereas double inhibition suppressed it (Figure 2E). The expression of other pro-inflammatory markers *TNF* and *IL1B* was induced by one of the inhibitors (Figure S2D). Interestingly, double inhibition induced anti-inflammatory cytokine *IL10* expression and secretion (Figure 2F) upon pro-inflammatory activation. IL-10 is known to be triggered by pro-inflammatory stimulation as a negative feedback loop to limit excessive inflammation.⁴² This could explain the reduced inflammatory response in double inhibitor-treated macrophages. Since some Trp derivatives can activate the aryl hydrocarbon receptor (AHR),^{43,44} we also examined the mRNA levels of *AHR* and its downstream responding genes *CYP1A1* and *CYP1B1*, which showed no significant differences (Figure S2F). These results indicate that while single blockage of Trp catabolism amplifies inflammatory responses, double inhibition leads to an anti-inflammatory effect, underscoring the complex role of Trp catabolic pathways in the pro-/anti-inflammatory phenotype in macrophages.^{45–47}

Integrated metabolic module-gene correlation uncovers an immunometabolic connection associating FcR signaling with glycolysis

To explore how environmental cues orchestrate metabolic modulation in macrophages, we applied the weighted gene correlation network analysis (WGCNA) algorithm^{48,49} to the combined metabolome (STAR Methods; Figure S3) and identified 65 highly interconnected modules (Table S1). Each module contains ions strongly correlated within the module across all macrophage subsets, representing a specific metabolic alteration that can be induced by one or more stimuli, even simultaneously by both pro- and anti-inflammatory activation. Next, we correlated the modules to the stimuli and performed MSEA for each module (Figure 3) and observed some patterns of correlation to specific activation subsets. For instance, module darkred, differentially correlated with pro- (positively) and anti-inflammatory (negatively) macrophages, consists of Trp catabolites such as nicotinate, QA, and KYN (Table S1), echoing the crucial role of Trp metabolism in macrophage inflammatory response. M-IL-4-specific modules (honeydew1, darkmagenta, turquoise, and red) were enriched in pathways related to lipid and fatty acid metabolism. This agrees with previous studies showing that lipolysis and fatty acid oxidation contribute to the IL-4-induced macrophage anti-inflammatory phenotype.^{50–52}

Subsequently, we conducted a thorough exploration of metabolic alterations by correlating metabolite modules with transcriptomic data (Star Methods; Tables S2 and S3). To validate

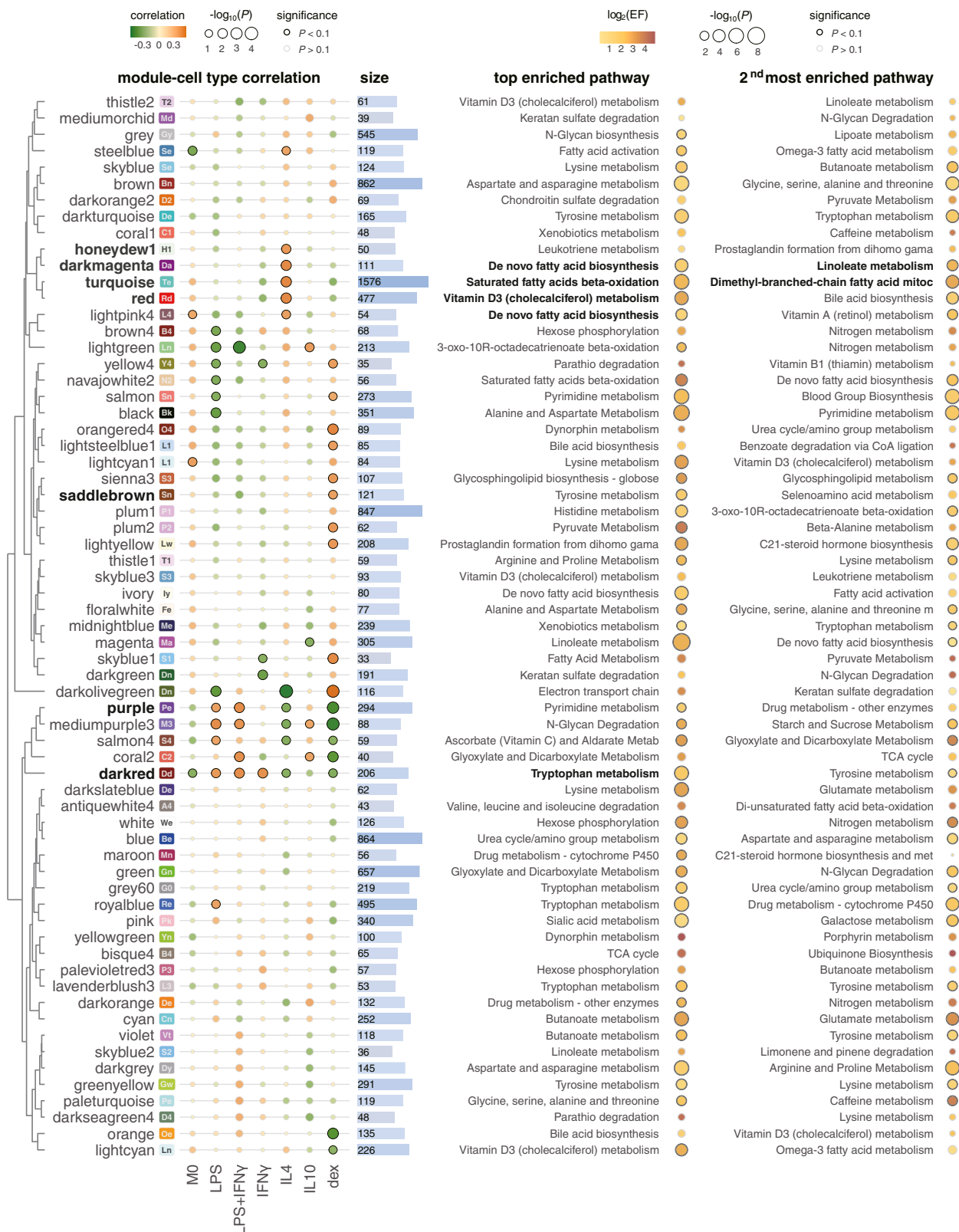
the robustness of this approach, we first focused on genes involved in key energy-generating processes: glycolysis, tricarboxylic acid (TCA) cycle, and glutaminolysis, along with their most-correlated modules (based on *p* values).²² This analysis revealed strong interconnections between catabolizing enzymes and key metabolites. For instance, module mediumpurple3, containing intracellular and supernatant lactate, showed the highest positive correlation with glycolysis enzymes (e.g., *HK1/3*, *PFKP*, *LDHA*) and a negative association with *MPC2*, the mitochondrial pyruvate carrier protein (Figure 4A). Module brown4, containing supernatant glucose, showed an inverse association with glucose transporter *SLC2A1* and glycolytic enzymes (e.g., *HK3*, *PKFM*). Module coral2, comprising intracellular phosphoenolpyruvate and α -ketoglutarate (α KG), displayed positive correlations with glycolytic enzymes and negative associations with genes downstream of α KG in the TCA cycle (e.g., *DLST*, *FH*, *MDH1*) and enzymes promoting glutamate- α KG conversion (*GOT1/2*). These findings confirm that the correlation between modules and gene expression accurately reflects the metabolic processes at play.

Next, we explored the potential to uncover potential immunometabolic regulation from the module-correlated genes that are not directly involved in metabolic processes. We performed Gene Ontology term analysis for each module and observed several known immunometabolic regulations, such as inverse correlations between supernatant glucose (module brown4) and pro-inflammatory pathways, as well as positive associations between intracellular glucose (module pink) and anti-pathogen responses (Figure S3). Notably, module mediumpurple3, representing enhanced glycolytic process (Figure 4A), showed a positive correlation with FcR-mediated signaling (Figure 4B). FcR signaling, together with Toll-like receptor (TLR) activation, induces an anti-inflammatory polarization in macrophages (M2b).^{41,53,54} As anti-inflammatory macrophages are considered to be less glycolysis dependent, we investigated whether FcR activation promotes lactate production. We generated M2b macrophages with immunoglobulin G immune complexes and either TLR3/4 agonist, poly(I:C), or LPS, respectively. As predicted by the module-gene correlation, we found that FcR activation triggered lactate production independent of TLR stimulation (Figure 4C). Furthermore, blocking Fc γ R signaling with Fc γ R neutralizing antibody mixture (Figure 4D) or an inhibitor against its downstream kinase SYK⁵⁵ (Figure 4E) eliminated immune complex-induced lactate production. These findings validate the ability of our model to predict immunometabolic reprogramming in human macrophages.

Figure 2. Suppression of inflammation through dual inhibition of tryptophan metabolism by IDO1 and IL411 inhibitors

(A–C) Metabolite levels and gene expression in Trp metabolism for each macrophage activation state. (A) Heatmap showing the scaled log-transformed average intensity of tentatively annotated intracellular and supernatant ions (with the *m/z* ratio in the bracket). (B) Heatmap showing the scaled log-transformed average of enzyme gene expression. (C) Schematic view of tryptophan metabolism (simplified from KEGG pathway hsa00380).

(D–F) qPCR and ELISA of pro-/anti-inflammatory marker genes in macrophages treated with IDO1 inhibitor (LRDS) and IL411 inhibitor (CB668) followed by pro-/anti-inflammatory stimuli (*n* = 4). (D) Anti-inflammatory *CCL22* gene and protein levels in IL-4-activated macrophages. (E) Pro-inflammatory *CXCL9* gene and protein levels with LPS plus IFN- γ stimulation. (F) Anti-inflammatory *IL-10* gene and protein levels with IL-4 or LPS plus IFN- γ stimulation. Data are represented as mean $\pm$ SD (*n* = 4). (*adjusted *p* < 0.05; **adjusted *p* < 0.01; ***adjusted *p* < 0.001; ****adjusted *p* < 0.0001; repeated-measures [RM] two-way ANOVA with Dunnett's multiple comparisons test, with a single pooled variance). Ctrl, control; fKYN, formyl-kynurenine; 3HAA, 3-hydroxyanthranilic acid; 3HK, 3-hydroxy-kynurenine; IP, indolepyruvate; KYA, kynurenine acid; NAD, nicotinamide adenine dinucleotide; NaMN, nicotinic acid mononucleotide.



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Stimuli-associated module network analysis reveals an inverse correlation between cholesterol accumulation and macrophage IFN- γ responses

Upon activation, metabolite levels in a metabolic process may change in opposing directions, with the product increasing and the reactant decreasing, and vice versa. Clusters of modules with inverse correlations can represent such metabolic alteration. To explore stimulus-specific module groups, we constructed a network considering both negative and positive inter-modular relationships (Figure S4A). For most macrophage activation states, stimulus-associated groups contained both negatively and positively interconnected modules (Figure S4B). Notably, a unique M-IFN γ -associated group (g2) consists solely of positively interconnected modules (Figure 5A), all of which showed a negative correlation with IFN- γ activation. Interestingly, this module group exhibited less association with other stimuli, including LPS+IFN- γ co-stimulation (Figure S4B). MSEA revealed an enrichment in cholesterol pathways of the intracellular metabolites, indicating an inverse relationship between intracellular cholesterol content and IFN- γ activation. To investigate the impact of cholesterol accumulation on macrophage response to IFN- γ , we treated MDMs with acetylated low-density lipoprotein (acLDL) overnight to generate cholesterol-loaded foamy macrophages, followed by IFN- γ activation (Figure 5C). Cholesterol accumulation was confirmed by lipid droplets (data not shown) and elevated cholesterol efflux genes *ABCA1* and *ABCG1* (Figure 5D). Cholesterol loading attenuated IFN- γ -responding genes, such as *CXCL9*, *IFIT1*, and *MX1* (Figure 5E), while other pro-inflammatory markers *TNF*, *IL6*, and *IL1B* remained unaltered or induced (Figure 5F). These data indicate that cholesterol accumulation specifically modulates IFN- γ response in macrophages.

DISCUSSION

In this study, we present a comprehensive multi-omics dataset to investigate the impact of metabolic processes on human macrophage function. By incorporating intra- and extracellular metabolomes with transcriptomic data, we construct a correlation network that characterizes macrophage activation. Through this integrated approach, we identify several immunometabolic modulations. Our findings reveal that simultaneous inhibition of tryptophan catabolism by IDO1 and IL4I1 inhibitors dampens pro-inflammatory macrophage features, while single inhibition leads to a more pro-inflammatory response. We also discover that M2b activation triggered by FcR signaling promotes glycolysis, contrary to the conventional notion of reduced glycolysis in anti-inflammatory activation. Additionally, we demonstrate that cholesterol accumulation specifically suppresses macrophage IFN- γ responses. This multi-omics dataset and approach provide valuable resources for researchers studying macrophage immunometabolic characteristics.

Nontargeted metabolomics offers a distinct advantage in explorative, hypothesis-generating research by acquiring

comprehensive ion features (*m/z* ratios) across samples with relative quantification. These features can be matched to known metabolites in curated databases, facilitating downstream analysis and discovery. These ion features, regardless of curated annotation, can serve as biomarkers for specific diseases or cellular states, aiding in early disease detection, classification, and monitoring of therapeutic interventions. This approach surpasses the observational bias often encountered in targeted methods (the streetlight effect), illuminating the path to uncovering previously unknown biomarkers and associations.

To explore immunometabolic modulation in macrophages, researchers often use differential analysis, comparing a condition of interest to a control (usually unstimulated macrophages). Differential analysis has limitations, overlooking unique characteristics of M0 and hindering a systemic, unbiased evaluation of diverse stimuli. To overcome these limitations, WGCNA emerges as a powerful tool for deciphering metabolic differences relevant to phenotypic and clinical features.^{48,49} Importantly, WGCNA can identify biomarkers independently of existing knowledge. For instance, our study reveals an immunometabolic characteristic of M2b macrophages where FcR signaling induces glycolysis. Tissue-derived *ex vivo* macrophages from immune complex-associated disease (lupus nephritis) patients and mouse models exhibit glycolytic metabolic rewiring.⁵⁶ In addition to FcR signaling, our analysis shows that lactate (module medium-purple3) positively correlates with actin filament organization and neutrophil-associated pathways. These pathways overlap in genes regulating phagocytosis and cytoskeletal dynamics, suggesting that metabolites in module mediumpurple3 play a role in macrophage morphology. This aligns with findings that macrophage phagocytic morphodynamics is fueled by glycolysis rather than OXPHOS.^{57,58} This FcR-glycolysis correlation is based on macrophage activation states unrelated to the primary stimuli we used to construct the network (immune complexes, poly(I:C), or their combination). This suggests that our module-gene correlation can be generalized to diverse activations, not restricted to the recruited subsets, uncovering associated immunometabolic processes. Additionally, uncharacterized genes correlated with metabolite modules may possess undiscovered enzymatic activity involved in corresponding reactions, highlighting the potential of our data resources for revealing unknown immunometabolic modulation.

Our data show opposing changes in Trp catabolism between pro- and anti-inflammatory macrophages aligning with the role of Trp metabolism via IDO1 in inflammation in both *in vitro* and *ex vivo* human macrophages.^{59,60} However, the immunomodulating effect of IDO1 inhibition remains controversial,^{47,61–63} with some studies showing anti-inflammatory features in *Ido1*-deficient mice⁴⁷ and others reporting increased IFN responses.^{61,62} Without pro-inflammatory challenges, *Ido1*-deficient mice show no obvious immune alterations.⁴⁵ Furthermore, clinical trials with IDO1 inhibitors did not yield promising results.⁶⁴ An alternative Trp catabolic process involves IL4I1,⁶⁵ whose role in macrophage function remains unclear. Some

Figure 3. WGCNA of metabolome classified ions into modules with distinct accumulation patterns

Shown are the correlation to the stimuli, module size (33–1576 ions per module), and the top two enriched pathways (by metabolite enrichment analysis) for the 65 modules identified by WGCNA signed network algorithm, with a soft power of 19 and a merging threshold of 10%.

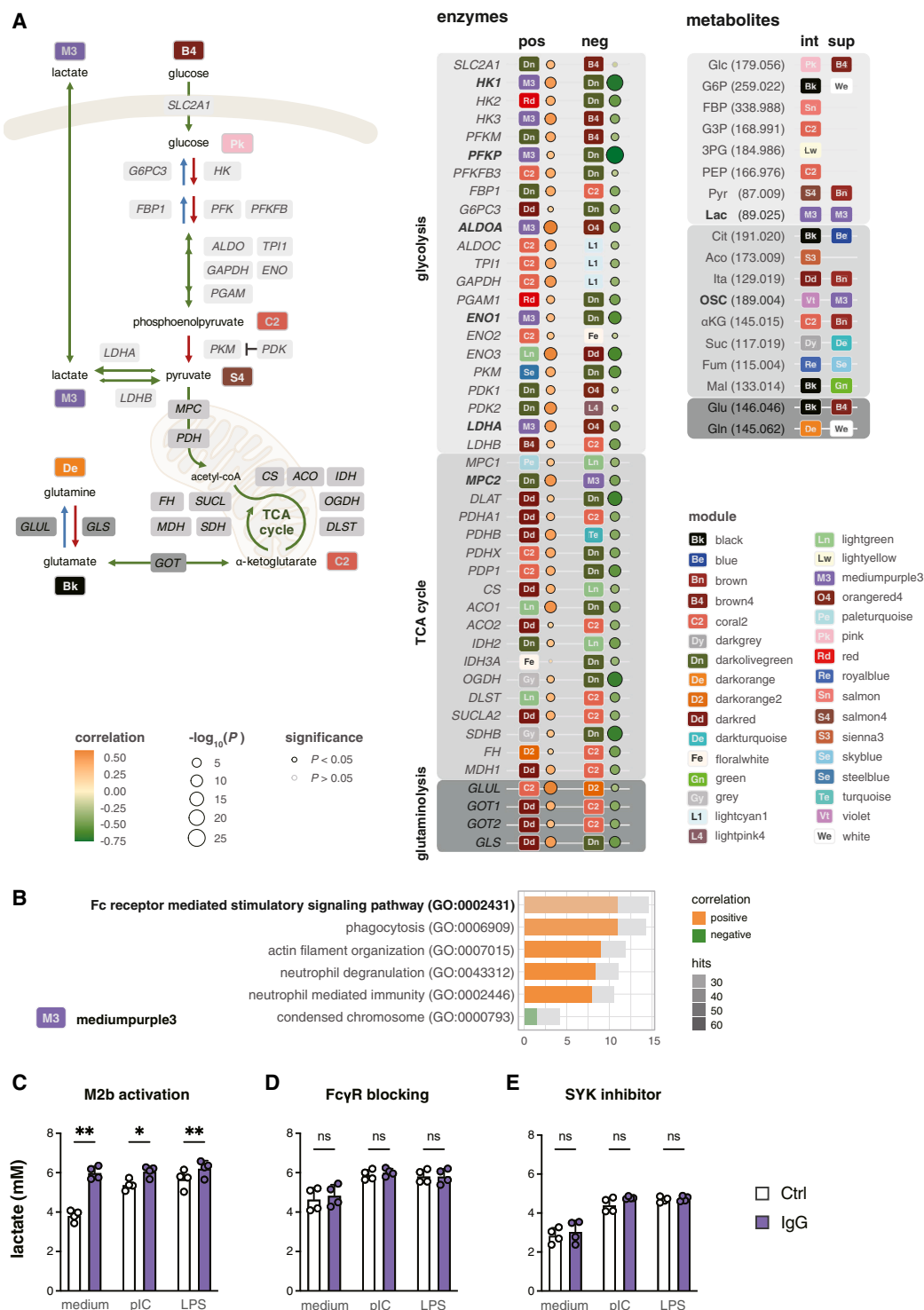


Figure 4. Integrated metabolic module-gene correlation uncovers an immunometabolic connection associating FcR signaling with glycolysis

(A) Intracellular (int) and supernatant (sup) ions and modules that are positively and negatively associated with genes involved in glycolysis, TCA cycle, and glutaminolysis (based on *p* values). The *m/z* ratio of the ions is indicated in the brackets.

(B) Top enriched pathways of the genes positively or negatively correlated with module mediumpurple3. Negatively correlated genes are enriched in only one pathway.

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studies suggest that *in vivo* IL411⁺ macrophages exhibit a strong pro-inflammatory program, while others propose immunosuppressive functions.^{66,67} Similar to *Ido1*-deficient mice, *Il4i1*^{−/−} mice do not exhibit an overt phenotype.⁶⁸ This may be because intestinal microorganisms can produce Trp derivatives downstream of IDO1 and IL411,⁶⁹ potentially contributing to the limited phenotypes observed in these *in vivo* mouse models. In the *in vitro* settings, inconsistent results across studies suggest that other pathways may compensate for the loss of IDO1 or IL411 alone. Our data bridge this gap by demonstrating that a complete blockage of Trp catabolism induces an anti-inflammatory effect using IDO1 and IL411 inhibitors. The potential mechanism may involve increased IL-10 production upon double inhibition, leading to an autocrine feedback loop that controls inflammatory responses. Notably, while *Ido1*^{−/−} mouse macrophages produce more IL-10,⁴⁷ we observed increased *IL10* expression in human macrophages only with double inhibition, not with the LRDS-only treatment. This discrepancy may be due to interspecies differences, inhibition efficiency between knockout and inhibitor, and culture conditions. Future studies investigating Trp in *Ido1* and *Il4i1* double-knockout mice are warranted.

Trp catabolites are highly associated with the anti-inflammatory metabolite itaconate,⁷⁰ with both belonging to module darkred (Figure S4; Table S1). Itaconate primarily originates from citrate, a TCA cycle intermediate, or it can be derived from glutaminolysis through IDH1.⁷¹ The significant correlation between module darkred and various TCA cycle and glutaminolysis genes (Figure 4) suggests a potential interplay between Trp metabolism, energy-generating processes, and itaconate production. Another Trp derivative, serotonin, has been shown to induce anti-inflammatory features in macrophages.^{72,73} Reduced serotonin levels caused by acute-phase IFN-driven inflammation are linked to long COVID pathophysiology.⁷⁴ Interestingly, intracellular serotonin (module saddlebrown) is positively associated with dexamethasone treatment, while supernatant serotonin (module purple) correlates positively with pro-inflammatory activation (LPS and LPS+IFN- γ) and negatively with anti-inflammatory subsets (IL-4, dexamethasone) (Figure 3). Whether IFN-macrophage crosstalk also plays a role in long COVID development warrants further investigation.

Furthermore, Trp catabolism contributes to NAD⁺ *de novo* synthesis, which is essential for maintaining the balance between glycolysis and mitochondrial respiration, as well as macrophage pro-/anti-inflammatory function.⁷⁵ Whether the effect of Trp metabolism modulation in macrophage function is related to downstream metabolites such as NAD⁺ or involves the regulation of associated processes like itaconate synthesis warrants further investigation.

Lipids also play crucial roles in coordinating macrophage phenotypes.⁵² In atherosclerosis, oxidized lipid species taken up by infiltrating monocytes and macrophages lead to chronic inflam-

mation in the arterial intima.⁷⁶ Intracellular cholesterol accumulation in these foamy cells has been linked to mitochondrial stress, inflammasome activation, and IL-1 β secretion.^{77,78} However, recent evidence from single-cell technologies suggests that atheroma-residing foamy macrophages exhibit very specific inflammatory features,^{79,80} and murine foam cell formation has been associated with suppressed inflammatory activation.⁸¹ Cholesterol-loaded macrophages show IFN hypo-responsiveness both *in vivo* and *ex vivo*. Monocytes from familial hypercholesterolemia patients accumulate lipids⁸² and show suppressed IFN responses, which are restored after cholesterol-lowering treatment.⁸³ Upstream regulator analysis indicates that STAT1 centers the regulatory network for this IFN hypo-responsiveness.^{83,84} Recent COVID-19 studies also support this correlation, showing weaker IFN responses in infected foam cells compared to infected non-foamy macrophages.⁸⁵ Similarly, lipid-loaded peritoneal macrophages from hyperlipidemic *ApoE*^{−/−} mice exhibit impaired IFN signaling and innate immune responses.^{83,86} Interestingly, the connection between IFN signaling and cellular cholesterol is bi-directional. IFNs reprogram cholesterol homeostasis in both mouse and human macrophages by decreasing cholesterol synthesis^{87,88} and stimulating the conversion of cholesterol to its derivatives.^{89,90}

In addition to our validated findings, our datasets have revealed other intriguing observations that could serve as valuable targets for future research. For instance, module darkolivegreen exhibits an inverse association with inflammatory mediators *TNF*, *CXCL9/10*, *IRF1/2/3/7/9*, and *STAT1/2*, while showing a positive correlation with anti-inflammatory markers, such as *MRC1*, *MARCO*, *ADORA3*, and *CD163L1* (Table S2). Furthermore, this module is closely related to key enzymes in glycolysis and the TCA cycle (Figure 4), suggesting the crucial roles of its member metabolites in regulating glucose flux and anti-inflammatory phenotypes in human macrophages. Notably, a module darkolivegreen member, moxifloxacin, has been shown to dampen LPS-induced pro-inflammatory responses in mouse peritoneal macrophages.⁹¹ Similarly, module pink, positively correlated with anti-pathogen responses, contains intracellular glucose and essential amino acids like threonine, isoleucine, and methionine. These amino acids are glucogenic and can fuel the TCA cycle via the succinyl-coenzyme A node. The accumulation of these metabolites implies a possible link between the inflammatory response and an alternative carbon source for energy in addition to amino acids. These insights highlight the potential significance of the identified modules in influencing macrophage function and warrant further studies.

In this paper, we established comprehensive multi-omics datasets of the non-targeted metabolome (intracellular and supernatant; Table S1) and transcriptome (Table S6) from seven commonly used human *in vitro* macrophage models. Based on these datasets, we created several key resources: metabolite module-gene correlation matrices (Tables S2 and S3) and

(C–E) Supernatant lactate concentration of (C) M2b, (D) M2b treated with Fc γ R neutralizing antibody mixture, and (E) M2b treated with 1 μ M SYK inhibitor entospletinib prior to FcR and TLR activation.

Data are represented as mean $\pm$ SD ($n = 4$) (*adjusted $p < 0.05$; **adjusted $p < 0.01$, RM two-way ANOVA with Geisser-Greenhouse correction; donor-matched Sidak's multiple comparison test). α KG, α -ketoglutarate; FBP, fucose-1,6-bisphosphate; Fum, fumarate; G3P, glyceraldehyde 3 phosphate; G6P, glucose-6-phosphate; Glc, glucose; Gln, glutamine; Glu, glutamate; Mal, malate; OSC, oxalosuccinic acid; 3PG, 3-phosphoglyceric acid; pIC, poly(I,C); Suc, succinate.

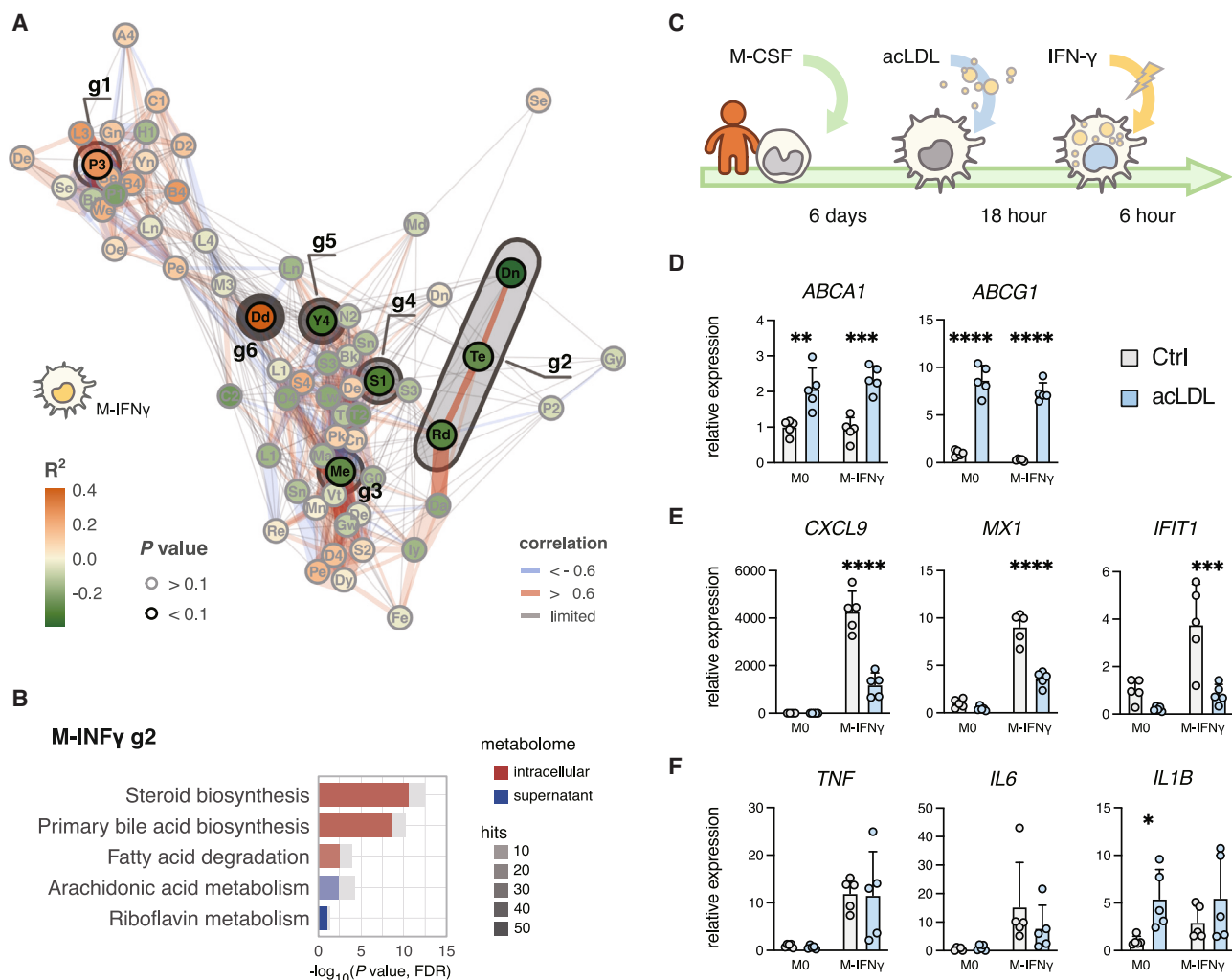


Figure 5. Module network reveals negative association between intracellular cholesterol and IFN- γ activation

(A) Signed module network displaying correlations between M-IFN γ and modules. The color of each module indicates the correlation coefficient (R) of the module eigen metabolite and IFN- γ treatment. The edge color and thickness represent correlation direction and absolute coefficient. Modules connected with an absolute $R > 0.5$ were grouped.

(B) MSEA on ions belonging to module cluster g2.

(C) Schematic experiment procedure: monocytes were differentiated into macrophages and loaded with acLDL, followed by IFN- γ treatment or remaining untreated.

(D–F) qPCR results of (D) cholesterol efflux genes, (E) IFN- γ -responding genes, and (F) pro-inflammatory cytokines.

Data are represented as mean $\pm$ SD ($n = 5$) (*adjusted $p < 0.05$; **adjusted $p < 0.01$; ***adjusted $p < 0.001$; ****adjusted $p < 0.0001$; RM two-way ANOVA; mixed effects models with donor-matched Sidak's multiple comparisons test).

module-subset/module-module correlation matrices (Tables S4 and S5). Future studies can leverage these resources in the following ways:

- (1) Studying similar macrophage subsets: for experiments focusing on similar macrophage subsets without a control (i.e., non-activated M0), our subset-correlated modules offer direct insights into stimulus-specific metabolite groups (Figures 3 and S3; Table S4).
- (2) Comparing subsets: for studies comparing two or more included subsets, researchers can use the established subset-correlated modules directly, as in approach 1, or

conduct differential analysis of our metabolomics/transcriptomics to identify more specific metabolic/transcriptional alterations.

- (3) Exploring other macrophage subsets: for research on macrophage subsets not included in this manuscript (e.g., other models, *in vivo* macrophages), the correlation matrices can be used to identify associated metabolite modules or genes.

a. Transcriptomic profile analysis: using gene sets specific to subsets, diseases, or pathways of interests, researchers can refer to Table S2 to find associated modules by weighted ranking or overrepresentation

analysis. Subsequently, the immunometabolic regulation network can be further explored using MetaboAnalyst (<http://www.metaboanalyst.ca>) or by mapping metabolites (and gene expression) to the Kyoto Encyclopedia of Genes and Genomes (KEGG) mapper (<http://www.genome.jp/kegg/mapper>).

- b. Metabolic profile analysis: for studies with metabolic profiles, overrepresentation analysis can identify the most enriched modules and their associated genes (Table S3). If only a few targeted metabolite levels are available, then researchers can identify associated pathways using Figure S3 and Table S3 without additional enrichment analysis.

We recommend investigating both combining or separating metabolites that are up- or down-regulated, as metabolic processes may show opposing changes in metabolite levels (with products increasing and reactants decreasing). As we provided both intracellular and supernatant metabolome data, our datasets can help interpret macrophage consumption or production of specific metabolites. In addition, we recruited gene expression profiles of short-term LPS-activated macrophages (3-h LPS treatment) for studying time-dependent gene expression alteration. These resources and approaches provide a comprehensive framework for utilizing our study in future immunometabolic research.

Notably, the majority of immunometabolic studies to date have been performed in mouse settings.⁹² While murine data support our findings regarding cholesterol-IFN and FcR-glycolysis, significant interspecies differences exist between mouse and human macrophage immunometabolism. For instance, LPS activation decreases OXPHOS in mouse macrophages, whereas human macrophages continue to utilize OXPHOS.^{93,94} This may explain why, in our correlation model, lactate is primarily associated with FcR signaling and actin reorganization rather than inflammatory pathways in human macrophages. Other key interspecies differences include the production of antimicrobial nitric oxide (NO). Mouse macrophages exhibit a robust NO response to inflammatory triggers both *in vivo* and *in vitro*, but NO production is often undetectable in many *in vitro* human macrophage models.⁹⁵ Our transcriptomic data also showed very low expression levels of NO synthase NOS2, which was filtered out during analysis, in all recruited human macrophage subsets. NO production is important in affecting macrophage plasticity in mouse macrophage re-polarization models.²¹ Therefore, it is important to consider the potential effects of NO when applying our findings to murine settings.

To conclude, we conducted gene-metabolite analysis in a diverse range of human macrophages, expanding beyond the limitations of mouse models⁹² and intracellular metabolome. We demonstrated that our resource datasets, combining metabolome and transcriptome, allow researchers to discover unknown macrophage immunometabolic characteristics. This comprehensive framework offers valuable information to unravel metabolic mechanisms underlying macrophage functional switches and provides potential avenues for therapeutic interventions in various macrophage-related disorders.

Limitations of the study

This study employs nontargeted metabolomics to acquire ion features (*m/z* ratios) with relative quantification. This approach has inherent limitations, such as ambiguous ion annotations and the inability to perform post-MS normalization due to its non-linear nature.^{96,97} Validation of the observed metabolic alterations will require confirmation through targeted metabolomic approaches or enzymatic reaction assays. Additionally, this work focuses on *in vitro* models and human primary monocyte-derived macrophages, and further validation is needed to confirm these effects in murine models and *in vivo* studies.

EXPERIMENTAL PROCEDURES

Details regarding the experimental procedures can be found in the [supplemental experimental procedures](#).

RESOURCE AVAILABILITY

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Materials availability

No new unique reagents or materials are generated in this study.

Data and code availability

- Original data: The transcriptomic data is available in the ArrayExpress repository under accession number E-MTAB-7572,¹⁵ with raw counts provided in Table S6. The intra- and extra-cellular metabolome data are shared on Figshare (<https://doi.org/10.6084/m9.figshare.26779603.v1>) and detailed in Table S1.
- Original code: This paper does not report original code. R packages and parameters used are stated in the [quantification and statistical analysis](#) section.
- Additional information: Other data that support the findings of this study are available from the corresponding author upon reasonable request.

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AUTHOR CONTRIBUTIONS

H.-J.C. performed the majority of the experiments, sample preparation, and analyses and drafted the manuscript. H.-J.C., G.R.G., and C.P.A.v.R. performed the validation experiments. H.-J.C. and J.V. performed the MS. D.C.S. developed the data analysis software and algorithms for MS. H.-J.C. and M.P.J.d.W. conceived the study and participated with L.M.B., B.S.W., P.G., P.K.M., D.C.S., J.d.D., R.K.P., and J.K. in the design and coordination of the study. M.P.J.d.W., W.J.d.J., and RKP provided the funding. All authors read and approved the final manuscript.

DECLARATION OF INTERESTS

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STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti-FcγRI	Abcam	CD64; clone 10.1; RRID: AB_3608287
anti-FcγRIIa	STEMCELL	CD32a; clone IV.3; RRID: AB_2925215
anti-FcγRIII	Abcam	CD16; clone 3G8; RRID: AB_3608288
Chemicals, peptides, and recombinant proteins		
Human M-CSF	Miltenyi Biotec	130-096-491
Lipopolysaccharides from Escherichia coli O55:B5	Sigma-Aldrich	L2637
Recombinant Human IFN-γ Protein	R&D	285-IF-100/CF
Recombinant Human IL-4	PeptoTech	200-04
Recombinant Human IL-10 Protein	R&D	217-IL-025/CF
Dexamethasone	Sigma-Aldrich	D4902
polyinosinic:polycytidylic acid (poly(I:C))	Sigma-Aldrich	P1530
Entospletinib (GS-9973)	MedChemExpress	HY-15968
Low-Density Lipoprotein from Human Plasma, Acetylated (acLDL)	Invitrogen	L35354
Critical commercial assays		
CD14 MicroBeads, human	Miltenyi Biotec	130-050-201
Human CCL22/MDC DuoSet ELISA	R&D	DY336
Human CXCL9/MIG DuoSet ELISA	R&D	DY392
Human IL-10 DuoSet ELISA	R&D	DY217B
Deposited data		
Transcriptome	EGA/ArrayExpress	ArrayExpress: E-MTAB-7572
Intra- and extra-cellular metabolome	Figshare	Figshare: 26779603
Software and algorithms		
GraphPad Prism version 9.5.1	GraphPad Software	N/A
R (v.4.0.5)	R Core Team	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human monocyte-derived macrophage (MDM) differentiation and stimulation

Human MDMs were generated following established protocols.¹⁵ In short, PBMCs were isolated from leukoreduction system chambers and buffy coats obtained with written informed consent from healthy anonymous blood donors in the UK and The Netherlands, respectively. CD14⁺ monocytes were isolated using CD14 magnetic beads and MACS columns (Miltenyi). The subsequent monocytes were then differentiated into macrophages over 6 days using 50 ng/mL human M-CSF (Miltenyi) and cultured in Iscove's Modified Dulbecco's Medium (IMDM) supplemented with 10% heat-inactivated fetal bovine serum, 1% Penicillin/Streptomycin solution, and 1% L-glutamine solution (Gibco). On day 6, the medium was replaced with culture medium without M-CSF and supplemented with the following stimuli: no stimulus, 10 ng/mL LPS (Sigma, E. coli E55:O5), 10 ng/mL LPS plus 50 ng/mL IFN-γ (R&D), 50 ng/mL IFN-γ (R&D), 50 ng/mL IL-4 (PeptoTech), 50 ng/mL IL-10 (R&D), 100 nM dexamethasone (Sigma) for 24 h. For metabolomics analysis, supernatants were collected, while intracellular metabolome samples were extracted with a 70% ethanol solution. Linrodostat (BMS-986205, 50 nM; MedChemExpress),³⁸ CB668 (10 μM; Calithera),³⁹ or dimethyl sulfoxide (DMSO, Sigma-Aldrich) was applied 1 h before the indicated stimulation. For M2b activation, MDMs were stimulated with high-affinity plate-immobilized human pooled serum IgG (Nanogam, 2 μg/mL), with or without TLR3/4 agonist poly(I:C) (20 μg/mL; Sigma-Aldrich), LPS. SYK inhibitor entospletinib (1 μM; MedChemExpress)⁵⁵ and FcγR neutralizing antibody mixture⁸ were applied 30 min before M2b activation, as previously described.^{8,55} For foamy macrophage formation,⁸³ acetylated LDL (50 μg/mL; Thermofisher) was loaded overnight (18 h), followed by IFN-γ activation. The chosen concentrations for the inhibitors (linrodostat, CB668, entospletinib) were determined based on optimal responses with no off-target cell killing effects observed.^{38,39,55}

Ethical declaration

The human studies presented in this work were approved by Amsterdam UMC Ethical Committee. All human biological samples were sourced ethically and their research use was in accord with the terms of the informed consent under an IRB/EC approved protocol.

METHOD DETAILS

Nontargeted metabolomics

To obtain metabolic profiles, we performed a non-targeted, flow-injection-analysis mass spectrometry method as previously described.^{98,99} In short, samples were analyzed on Q-exactive Plus (Thermo Scientific) with a heated electrospray ion source and operated in negative ionization profile mode with scan range 50–1000 m/z. Resolution was set to 70,000 at 200 m/z with automatic gain control target of 3E6 ions, 3.0 kV spray voltage, 120 m maximum injection time, and 160 s acquisition time. Samples were injected in a randomized sequence and analyzed using a mobile phase consisting of 60:40% (v/v) isopropanol-water buffered with 1 mM ammonium fluoride at pH 9 and containing 10 nM taurocholate and 20 nM homotaurine. Three types of quality control samples were analyzed every 20 injections: solvent blanks to detect and filter noise, a mixture of seventeen defined standards to ensure consistent instrument performance, and a mixture of aliquots of all samples to compensate for instrument drift on an ion level. Spectral data processing was performed as described previously,^{98,99} with a signal-to-noise threshold of 2.0. Detected ion m/z values and isotope distributions were matched against the Human Metabolome Database¹⁰⁰ based on accurate mass within a tolerance of 5 mDa assuming [M-H] and [M-2H] species and at most two ¹³C/¹²C exchanges to tentatively annotate metabolites, with the inherent limitation of being unable to distinguish between isomers. No additional normalization was conducted due to the method's non-linear nature.

Quantitative polymerase chain reaction

Total RNA was isolated with the RNeasy Mini Kit (Qiagen) as per the manufacturer's protocol. The resulting RNA was then converted to complementary DNA with iScript (Life Technologies). qPCR was performed with SYBR Green Fast on a ViiA7 PCR machine (Applied Biosystems). Data were normalized to the mean of the relative expression values of two housekeeping genes, *HPRT1* and *RACK1*. Primers are as follows:

HPRT1 (FW: GACCAGTCAACAGGGGACAT, RV: AACACTTCGTGGGGTCCTTTTC)
RACK1 (FW: GAGTGTGGCCTTCTCCTCTG, RV: GCTTGACAGTTAGCCAGGTTTC)
ABCA1 (FW: CCTGTCATCTACTGGCTCTC, RW: ACAACGTAATTGCACATATCCC)
ABCG1 (FW: GAGATGGGAGTCTTTCTTCGG, RV: CACTGGGAACATGATCTGAAAGG)
TNF (FW: GGCGTGGAGCTGAGAGAT, RW: TGGTAGGAGACGGCGATG)
CCL22 (FW: CCTACTCTGATGACCGTGGC, RV: AGGGAATGCAGAGAGTTGGC)
IFIT1 (FW: AAGCTTGAGCCTCCTTGGGTTCGT, RV: TCAAAGTCAGCAGCCAGTCTCAGG)
MX1 (FW: GGCATAACCAGAGTGGCTGT, RV: CATTACTGGGGACCACCACC)
CXCL9 (FW: GACCTTAAACAATTTGCCCAAG, RV: TCCTTCACCCCCATCTGCTGAATCTGG)
IL1B (FW: AAGATGCTGGTTCCTCTGC, RV: GTTCAGTGATCGTACAGGTGC)
IL6 (FW: GAGTAGTGAGGAACAAGCCAG, RV: TTGTCATGTCCTGCAGCC)
AHR (FW: CAAATCCTTCCAAGCGGCATA, RV: CGCTGAGCCTAAGAACTGAAAG)
IL10 (FW: GGGAGAACCTGAAGACCCTC, RV: AAGAAATCGATGACAGCGCC)
CYP1A1: (FW: TCGGCCACGGAGTTTCTTC, RV: GGTCAGCATGTGCCCAATCA)
CYP1B1: (FW: GCCACTATCACTGACATCTTCGG, RE: CACGACCTGATCCAATTCTGCC)
MRC1 (FW: GCCAACACAGAACGCTGAG, RV: CTGCCCAGTACCCATCCTTG)
CD209 (FW: AATGGCTGGAACGACGACAAA, RV: CAGGAGGCTGCGGACTTTTT)

Enzyme-linked immunosorbent assay (ELISA)

Supernatants were harvested after 24 h of stimulation, and cytokines were detected using the specified antibody pairs according to the manufacturer's protocol: Human CCL22/MDC DuoSet ELISA (DY336, R&D), Human CXCL9/MIG DuoSet ELISA (DY392, R&D), Human IL-10 DuoSet ELISA (DY217B, R&D).

Lactic acid measurement

Lactic acid in the supernatant was measured as previously described.¹⁰¹ In short, supernatant was first harvested and deproteinized by 3% (final concentration, w/v) metaphosphoric acid. Lactate content was then assayed by lactate dehydrogenase (Roche)¹⁰² followed by measurement with CLARIOstar microplate reader (BMG).

QUANTIFICATION AND STATISTICAL ANALYSIS

Metabolomic data preprocessing

After combining intracellular and supernatant metabolomes to one single data matrix and filtering out low-intensity ions ($<10^5$ AU, 2514 ions removed), we applied random forest imputation^{30,31} for ions containing less than 33.3% missing values for downstream analyses (Figure 1B). Ions with missing values more than this threshold were removed from the data matrix (1606 ions). In total 28623 out of 563514 values (5.1%) were imputed based on random forest algorithm. After subsequent filtering with goodSamples-Genes function (4120 ions removed) from R package WGCNA,⁴⁸ the final combined intra- and extracellular metabolome resulted in 13418 ions (Table S1), of which 5136 were putatively mapped to known metabolites based on their mass-to-charge features (m/z ratios).

Missing value imputation

Simply removing missing values or replacing with determined values beforehand for statistical analysis could lead to bias.^{49,103} Missing value imputation is applied primarily for analysis most robustly without missing content, such as principal component analysis (PCA) and partial least squares-discriminant analysis (PLS-DA).¹⁰⁴ Random forest is a classification and regression algorithm capable of handling both parametric and non-parametric datasets of complex linear and non-linear.³⁰ RF imputation has been shown to be favored for metabolomic studies over other methods.^{103,105,106} Thus, to decrease the risk of removing potential important metabolites, miss value imputation was performed with RF algorithm by R package missForest v.1.4³¹ after removing ions of which the proportion of missing values were accounted for more than 33.3%. After imputation and filtering, we performed PCA and PLS-DA for both unimputed and imputed datasets which showed comparable results revealing similar enriched pathways (Figure S6).

Weighted gene correlation network analysis (WGCNA)

The R package WGCNA⁴⁸ was used to investigate the metabolomic networks based on ions' correlation patterns. Directly associated ions were clustered together using biweight midcorrelation, dynamic tree cut algorithm and signed networks. The soft threshold was set to 19, which is the lowest power for which the scale-free topology fit index (R^2) reaches 0.90 (Figure S3A). In the hybrid dynamic tree cutting algorithm, deepSplit was set to 4 and the minimum module size was set to 30. Modules sharing correlation with each other at 90% or greater were merged. For the downstream analyses, we included all ion features passed the quality controls. As if only annotated metabolites were recruited in clustering approach, metabolites might be assigned to suboptimal module due to missed connections between the unassigned ions. For the resulting modules, a summarizing eigen values was then calculated based the first principal component. Subsequent associations between module eigen values with activation traits were computed using biweight midcorrelation. Modules connected with an absolute $R > 0.65$ were grouped for downstream analyses. For stimulus-associated group, a cut-off of 0.5/-0.5 between modules correlated with the stimulus was applied for grouping.

Transcriptomic data

The processed transcript counts were retrieved from the ArrayExpress repository E-MTAB-7572.¹⁵ For each donor, variance stabilization transformed expression data were assigned as a phenotype trait to all the metabolome samples with the same activation states. Correlation between the module eigen ions and the transcriptome traits was calculated with biweight midcorrelation. To obtain weighted summary correlation coefficient and the corresponding p values from three donors, we performed Fisher's z transformation followed by meta-analysis using Hedges and Olkin's fixed-effect method.¹⁰⁷

Pathway enrichment analysis

For the transcriptomic data, Gene Ontology term enrichment analysis was performed by using R package clusterProfiler v.4.0.¹⁰⁸ For metabolomic profiles, metabolite set enrichment analysis was performed with MetaboAnalystR v3.0 package.¹⁰⁹ Enrichment factor represents the ratio of the Q-statistic for each pathway to the expected statistic of the particular pathway.

Statistical analysis

With the exception of the qPCR data, all analyses were performed in R (v.4.0.5) environment. Graphs were created with R packages igraph, pheatmap, ggplot2, and ggpvr. For qPCR and ELISA data, analyses were conducted in GraphPad Prism v.9.5.1 (GraphPad Software). Data were first examined with D'Agostino-Pearson normality test with $\alpha = 0.05$. For multiple comparison tests, one-way or two-way analysis of variance (ANOVA) was applied on the basis of the addressed scientific question.